

NATIONAL TECHNICAL RESEARCH ORGANISATION

SIGNAL ANALYSIS & PARAMETER EXTRACTION DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: ad6f79d55dfd92e3

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_qpsk.iq
File SHA-256 Digest : ad6f79d55dfd92e3db8e1a8e4b6828a954c00c3e42091a45f8c4d8a2193216cb
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-18T19:00:35Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	2.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+122.394 kHz	Spectral Centroid Offset
Estimated Baud Rate	250.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	1.46 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	293.95 kHz	Cumulative PSD 99% Energy
Estimated SNR	9.23 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	4.06 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+99975.59 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0110	Wiener Entropy Index (0.0 – 1.0)

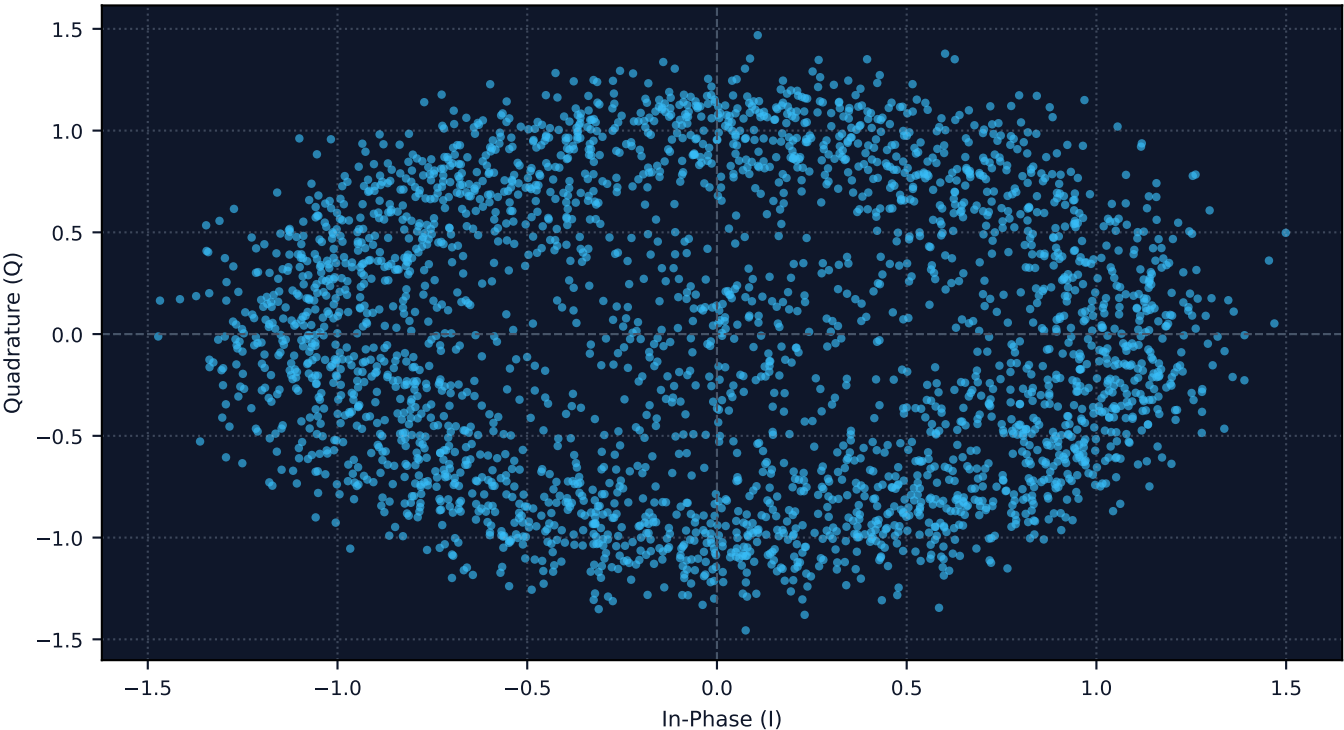
PREDICTED MODULATION : QPSK

CLASSIFIER CONFIDENCE : 76.94% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: QPSK | Recovered Bits: 131,072 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [QPSK]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	51	0x0000	0	0 B	CRC MISMATCH	CRC_MISMATCH
#2	371	0x0000	1	16330 B	CRC MISMATCH	PARTIAL_RECOVER
#3	1427	0x0000	2	16198 B	CRC MISMATCH	PARTIAL_RECOVER
#4	3555	0x0000	3	15932 B	CRC MISMATCH	PARTIAL_RECOVER
#5	3877	0x0000	4	15892 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

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[FRAME 1 PAYLOAD]